

Student Takeaway for Sync Session

<< Week 13: Regression Methods - I >>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Linear Regression Workshop	
Session duration	1Hr:45 minutes	
Session type	Lectures: Conceptual understanding of supervised learning, linear models, overfitting, and regularisation. Hands-on: Build, train, evaluate, and visualise linear, multiple linear, and polynomial regression models using Python.	
Scope	This session explores practical applications of regression in machine learning. It covers: - Core ML concepts and supervised learning - Linear, multiple, and polynomial regression - Dataset preparation, model training, and evaluation - Visualising predictions and interpreting model metrics	
Learning objectives	Objective	Core capability
	Understand supervised learning and linear regression concepts	Build foundational ML knowledge
	Apply linear, multiple linear, and polynomial regression	Hands-on experience with regression models
	Evaluate model performance	Use metrics like RMSE and R^2
	Use Python ML tools for implementation	Technical proficiency in scikit-learn and matplotlib
Software/Tools	- Python (Libraries: pandas, numpy, matplotlib, sklearn) - IDE: Jupyter Notebook or Colab - Dataset: house.csv - Presentation Tool: PowerPoint	

Pro tips for success:

- Focus on Fundamentals:** Understand how input features influence predictions.
- Be Visual:** Use regression plots to interpret model behavior.
- Think Critically:** When would a simple line not be enough? Explore non-linear options like polynomial regression.

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1-3	ML Concepts & Supervised Learning Parameter Estimation & Overfitting	Learn about input-output data mapping and linear models Learn least squares method and why overfitting happens	Foundation of predictive modeling Balance fit and complexity
4	Simple Regression Steps	Key steps: load data, train/test split, fit, predict	Hands-on Python walkthrough
5	Dataset Setup	Load and explore house pricing dataset using pandas	Use <code>df.head()</code> to inspect data
6-13	Linear Regression Setup step 1 - 8	Train model on Area→Price, plot results, check accuracy	Import libs, split data, fit model, evaluate with R^2 & RMSE, plot train/test lines
14-16	Make Predictions, Compare Predictions and Model Summary	Predict test data, plot actual vs predicted, show R^2 , slope, intercept	<code>predict()</code> , scatter plot, print scores
17	Why Polynomial Regression	Use when data isn't linear; better fits curves	Linear fails on curves; e.g., Age vs Income relationship
18-20	Multiple Linear Regression Predicting House Prices Required Libraries	Predict with multiple features, encode categories	Use dummies, fit model, check R^2 & RMSE
21-25	Multiple Linear Regression step 2 - 8	Encode, train, evaluate, plot	• Use dummies for categories • Split data, fit model • Check R^2 & RMSE • Plot actual vs predicted
26-27	Polynomial Regression	Fit curves with polynomial terms	Linear model with powers of x to capture non-linearity

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28	Why Polynomial Regression	For curved data, better fit than linear	• Easy, flexible • Risk overfitting Harder to interpret at high degrees
29	Polynomial Regression with PolynomialFeatures	Add polynomial terms for non-linear fit	Linear model, captures curves
30-31	Load required libraries	Create polynomial features, train model	Use PolynomialFeatures and LinearRegression
32-34	Polynomial Regression – Model	Check model accuracy on train/test sets	• Calculate RMSE and R^2 for train and test • Review results and discuss questions

Post session

Reflection challenge	What's the biggest difference between simple, multiple, and polynomial regression—and when would you use each?
Explore more	Try tweaking your model: • Add a new predictor (e.g., 'Rooms') • Change the polynomial degree • Compare models using evaluation metrics
Get inspired	Real-world pricing, forecasting, and trend analysis rely on regression—mastering this unlocks a key ML skillset!
The journey ahead	In our next session, we'll go deeper into classification models—understanding how to categorise and make decisions from data.